

MedPath India: Technical Design Document

AI-Powered Healthcare Navigator and Treatment Cost Estimator Platform

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Author: System Architecture Team

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1. Overview

1.1 Executive Summary

MedPath India is a real-time healthcare intelligence platform designed to help Indian patients—particularly those in Tier 2 and Tier 3 cities—discover optimal hospitals based on their medical condition, location, affordability, and personal context. The system translates patient intent expressed in natural language (Hindi, Hinglish, or English) into actionable healthcare recommendations with transparent cost estimates.

1.2 Problem Statement

Indian patients, especially in non-metro areas, face significant challenges:

- **Information asymmetry:** Difficulty finding appropriate hospitals for specific conditions
- **Cost uncertainty:** Unpredictable treatment costs and hidden charges
- **Language barriers:** Most health-tech solutions are English-first
- **Trust deficit:** No reliable way to assess hospital quality before visiting
- **Geographic constraints:** Limited awareness of accessible healthcare options

1.3 Solution Approach

MedPath India operates as a "Search-first + Intelligence-memory" architecture:

- **Real-time discovery:** Hospitals are discovered dynamically from public sources (Google Maps, Practo, JustDial, hospital websites, NABH records)
- **Intelligence memory:** The system stores learned intelligence (feedback, actual costs, trust signals) rather than maintaining a complete hospital database
- **Explainable recommendations:** Every recommendation includes transparent reasoning
- **Self-improving:** Feedback loops continuously improve ranking accuracy

1.4 Critical Disclaimer

This system is **decision-support only**. It does NOT:

- Diagnose diseases
- Replace medical professionals
- Provide medical advice
- Guarantee treatment outcomes

2. Goals and Non-Goals

2.1 Goals

Goal	Success Metric	Priority
Enable patients to discover relevant hospitals within 60 seconds	Time-to-first-recommendation < 60s	P0
Provide cost estimates within 20% accuracy of actual costs	Cost estimation accuracy measured via feedback	P0
Support Hindi, Hinglish, and English natural language input	Intent extraction accuracy > 85%	P0
Deliver explainable recommendations	User understanding score > 4/5 in surveys	P0
Improve recommendations through feedback	Recommendation acceptance rate improvement over time	P1

Support Tier 2/3 city users effectively	Equal or better experience metrics vs metro users	P1
Maintain sub-3-second response times for recommendations	P95 latency < 3s	P1

2.2 Non-Goals

Non-Goal	Rationale
Medical diagnosis	Regulatory and safety concerns; out of scope
Appointment booking	Phase 2 feature; requires hospital integrations
Insurance claim processing	Separate product domain
Telemedicine/video consultations	Different product category
Maintaining complete India hospital database	Search-first architecture preferred
Real-time bed availability	Requires hospital system integrations
Prescription management	Out of scope for navigator product

3. Product Architecture Overview

3.1 Core Architecture Philosophy

ARCHITECTURE PRINCIPLES

1. Search-First: Discover hospitals dynamically, don't maintain full DB
2. Intelligence-Memory: Store learnings, not raw hospital data
3. Explainability-First: Every recommendation must be justifiable
4. Graceful Degradation: System works even when external sources fail
5. Privacy-by-Design: Minimize PII storage, anonymize feedback
6. Modular Monolith: Start simple, extract services when needed

3.2 System Boundaries

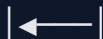
SYSTEM BOUNDARY

PATIENT



MEDPATH INDIA PLATFORM

(User)



EXTERNAL DATA SOURCES

(Google Maps, Practo, JustDial,

NABH, Hospital Websites)

OUT OF SCOPE: Hospital internal systems, Insurance systems,
Pharmacy systems, Lab systems

4. End-to-End System Workflow

4.1 Primary User Flow

END-TO-END USER JOURNEY

PHASE 1: INPUT CAPTURE

Step 1: User opens app

Step 2: User provides input (voice OR text)

Step 3: If voice → Speech-to-Text conversion

Step 4: Raw text captured in Hindi/Hinglish/English



PHASE 2: INTENT EXTRACTION

Step 5: NLU pipeline processes text

Step 6: Extract:

- Symptoms/medical condition
- Possible medical category
- Urgency level
- Patient context (if provided)

| Step 7: Identify missing MANDATORY information



PHASE 3: CONTEXT COMPLETION

| Step 8: Check mandatory context:

- | - Location (GPS or address) ← MANDATORY
- | - Medical issue/symptoms ← MANDATORY

| Step 9: If missing → Generate follow-up questions

| Step 10: Collect optional context if volunteered:

- | - Age, comorbidities, insurance, budget, hospital preference



PHASE 4: HOSPITAL DISCOVERY

| Step 11: Determine search radius based on:

- | - Urgency (emergency = smaller radius)
- | - City density (metro vs rural)
- | - Speciality rarity (rare = wider radius)

| Step 12: Query external sources in parallel:

- Google Maps API
- Practo scraper/API
- JustDial scraper
- NABH database lookup

Step 13: Aggregate and deduplicate results

|



PHASE 5: FILTERING

Step 14: Apply exclusion filters:

- Irrelevant speciality
- Poor review sentiment (< threshold)
- Extreme distance
- Weak emergency capability (if urgent)
- Affordability mismatch (if budget provided)
- Low trust score (from intelligence memory)

|



PHASE 6: ENRICHMENT

Step 15: Enrich filtered hospitals with intelligence memory:

- Past feedback scores
- Real treatment costs (from previous users)
- Hidden charge warnings
- Wait time estimates
- Trust signals

Step 16: Fetch real-time data where possible:

- Current reviews
- Operating hours
- Contact information



PHASE 7: RANKING

Step 17: Calculate composite score using weighted factors:

HIGH PRIORITY:

- Medical specialization relevance (weight: 25%)
- Hospital capability match (weight: 20%)

- Distance/travel time (weight: 15%)

- Affordability score (weight: 15%)

- Review sentiment (weight: 10%)

- Accreditation (weight: 10%)

- Historical feedback (weight: 5%)

MEDIUM PRIORITY (tie-breakers):

- Wait times, billing transparency, suitability flags

Step 18: Generate explanation for each rank



PHASE 8: COST ESTIMATION

Step 19: For top-ranked hospitals, generate cost estimates:

- Procedure cost

- Consultation fees

- Diagnostics

- Medicines (estimated)

- Hospital stay (if applicable)

- Contingency buffer

| Step 20: Apply adjustments for:

- | - City tier
- | - Hospital type (govt/private/trust)
- | - Patient comorbidities
- | - Historical actuals from feedback



PHASE 9: PRESENTATION

| Step 21: Render results as hospital cards:

- | - Name, ratings, distance, travel time
- | - Estimated cost range
- | - Key facilities
- | - "Why recommended" explanation
- | - Trust score, accreditation badges
- | - Action buttons (call, navigate)



PHASE 10: POST-TREATMENT FEEDBACK

Step 22: After treatment (triggered via notification):

- Actual treatment cost
- Hospital experience rating
- Doctor experience rating
- Hidden charges encountered
- Wait time experienced
- Billing transparency rating
- Recommendation likelihood (NPS)

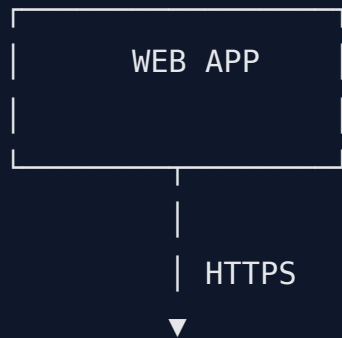
Step 23: Update intelligence memory

Step 24: Retrain/adjust ranking weights periodically

5. High-Level Architecture Diagram

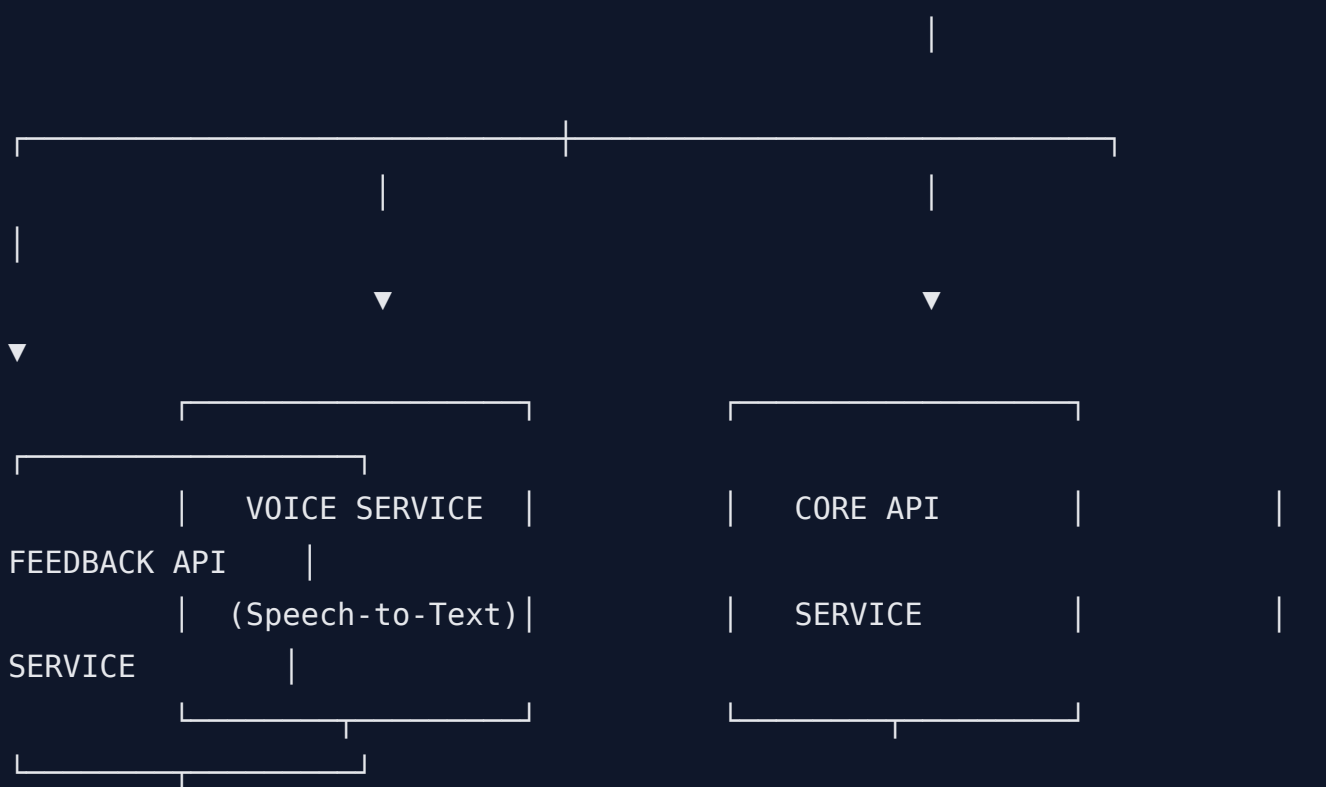
MEDPATH INDIA - HIGH LEVEL

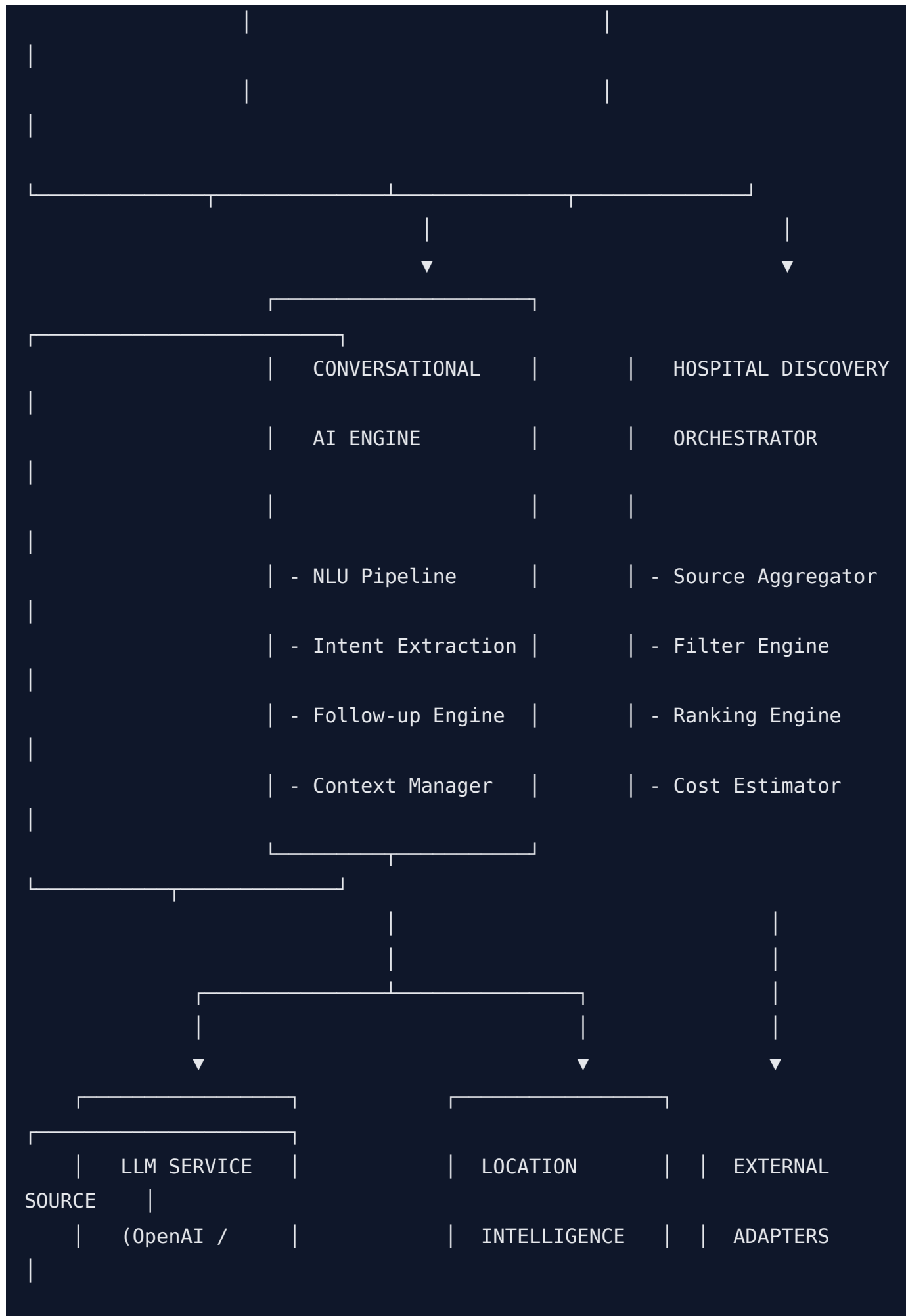
ARCHITECTURE



API GATEWAY / LOAD BALANCER

(nginx / AWS ALB)







COMPONENT ARCHITECTURE DETAIL

PRESENTATION LAYER

MOBILE APPLICATION

Voice

Feedback

Input

Form

Screen

Screen

Chat

Interface

Screen

Results

Display

Screen

Hospital

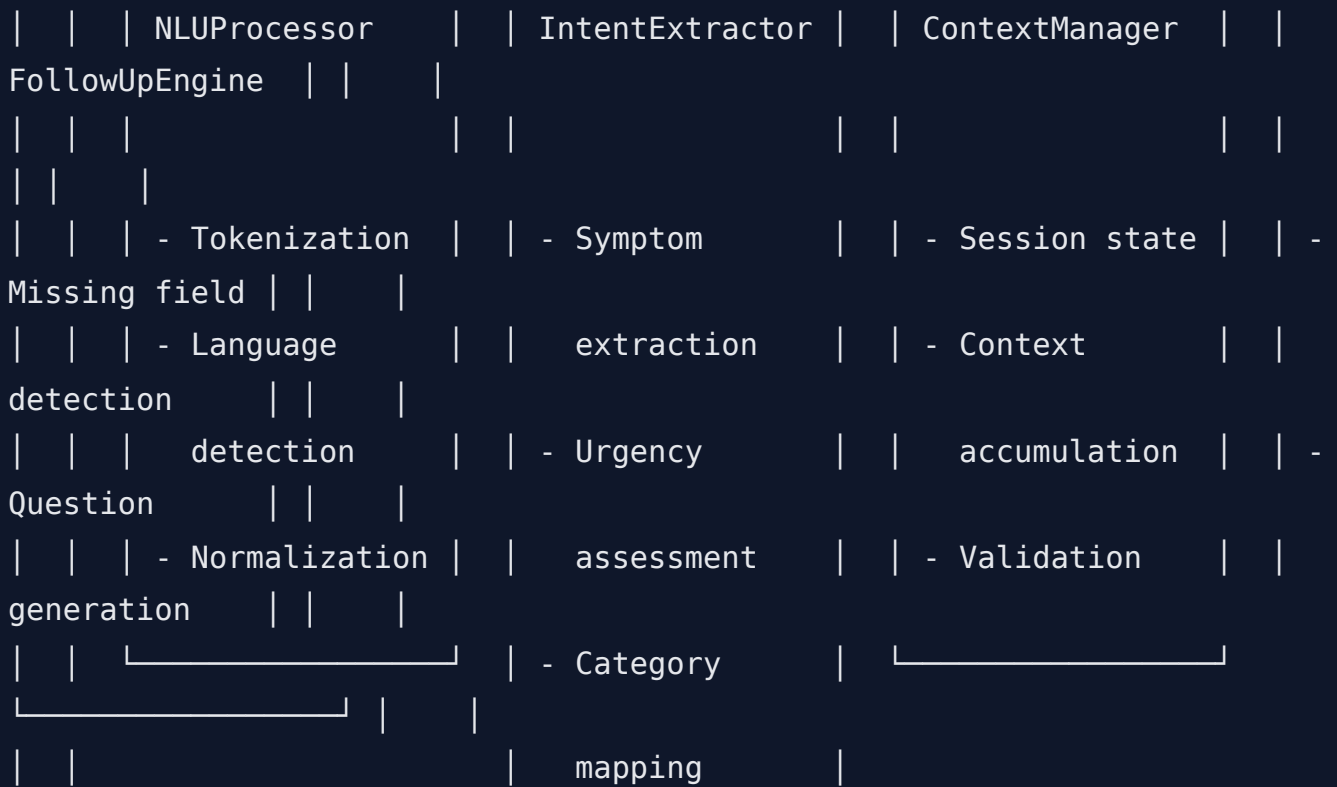
Detail

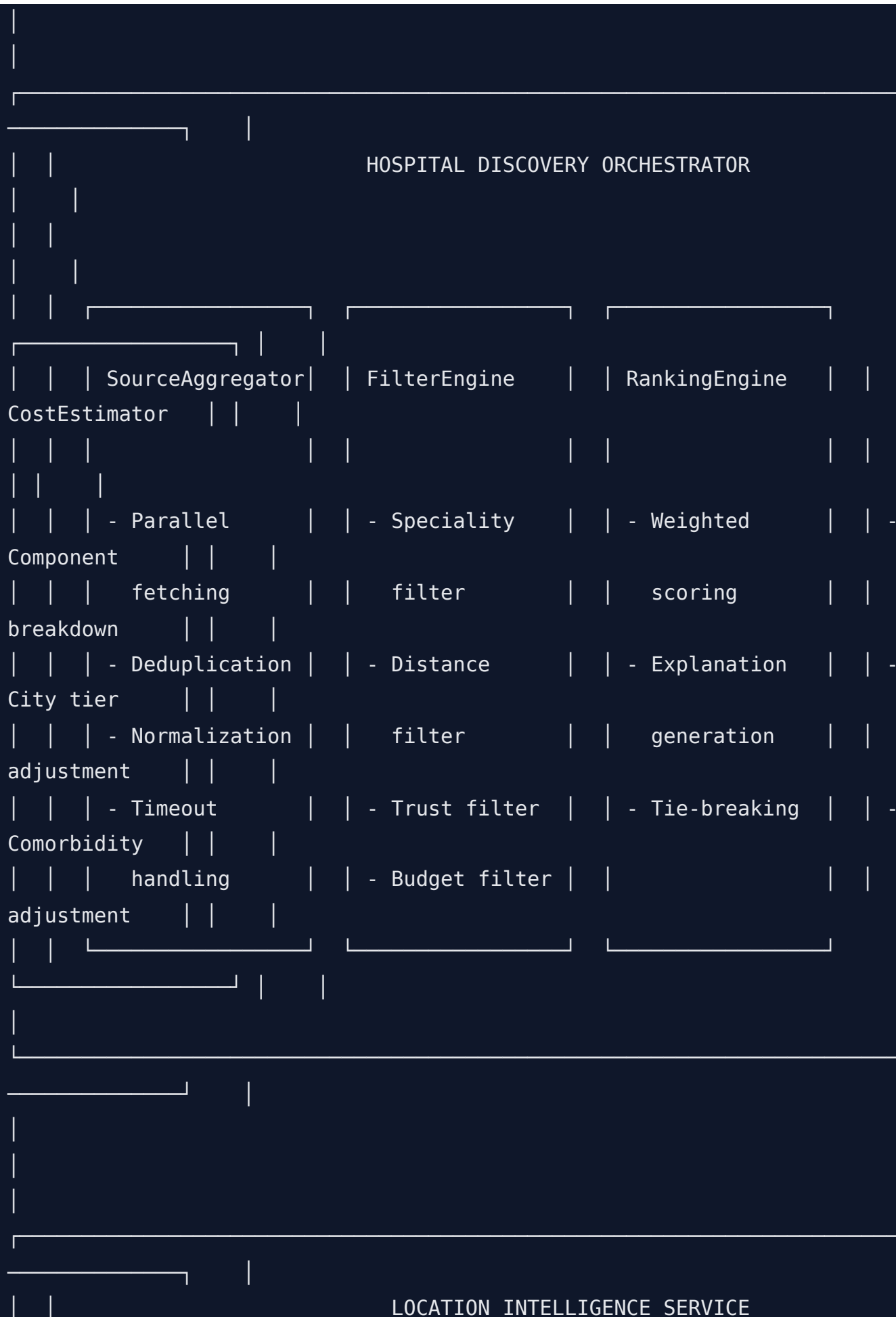
Screen



DOMAIN LAYER

CONVERSATIONAL AI ENGINE





GeocodingService	RadiusCalculator	TravelTimeCalc
- Address to coordinates	- Urgency-based radius	- Distance matrix
- GPS fallback	- City density adjustment	- Traffic estimation
- Validation		